

SMARTSHELF WEIGHT MEASUREMENT & INVENTORY TRACKING SYSTEM

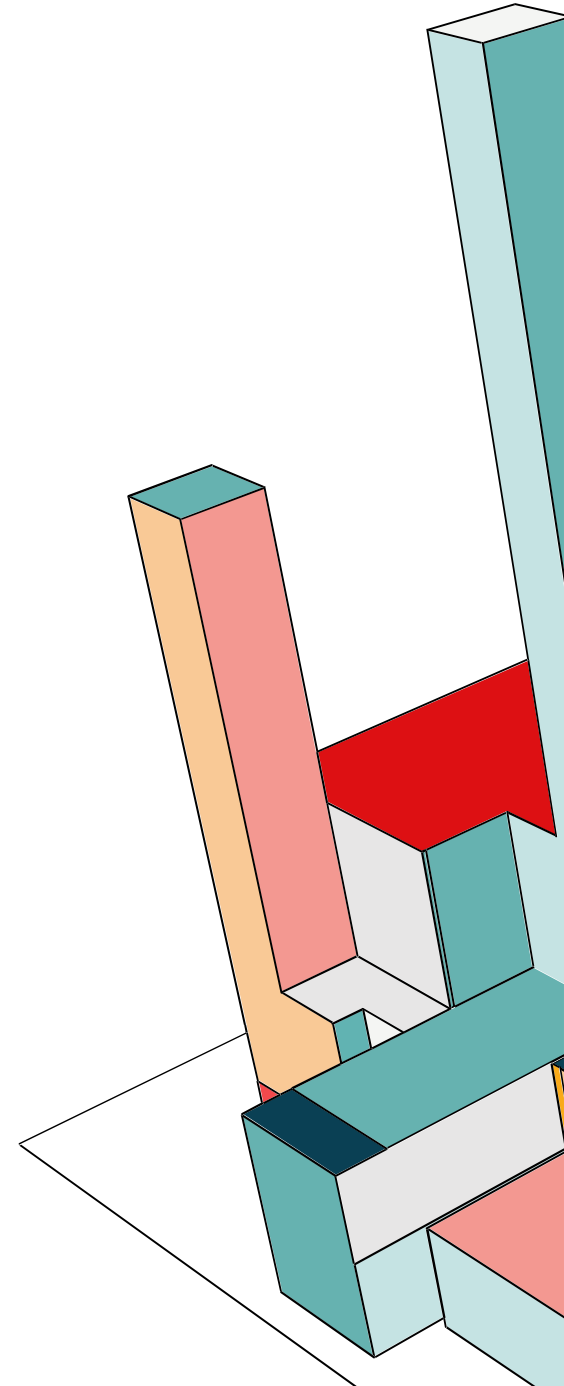
Abir Alasam

August 21, 2025



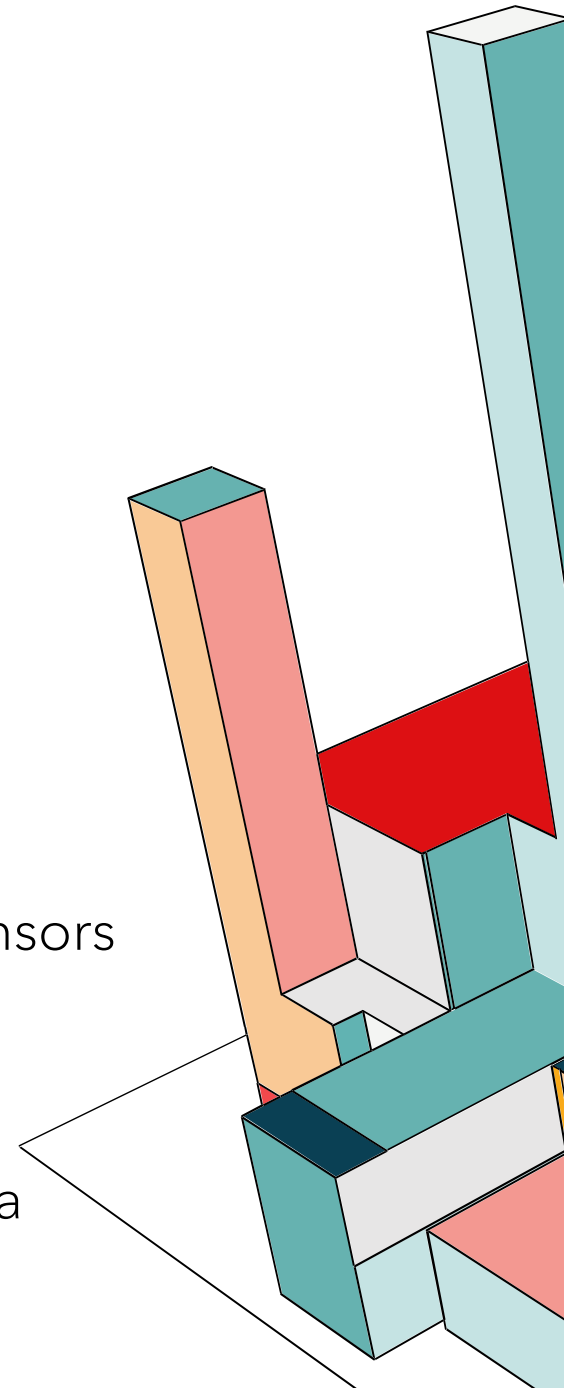
AGENDA

- Introduction
- System Overview
 - Hardware Setup
 - Software Development
 - Odoo Integration
- GUI & Visualization
- Challenges & Solutions
- Results & Achievements
- Conclusion

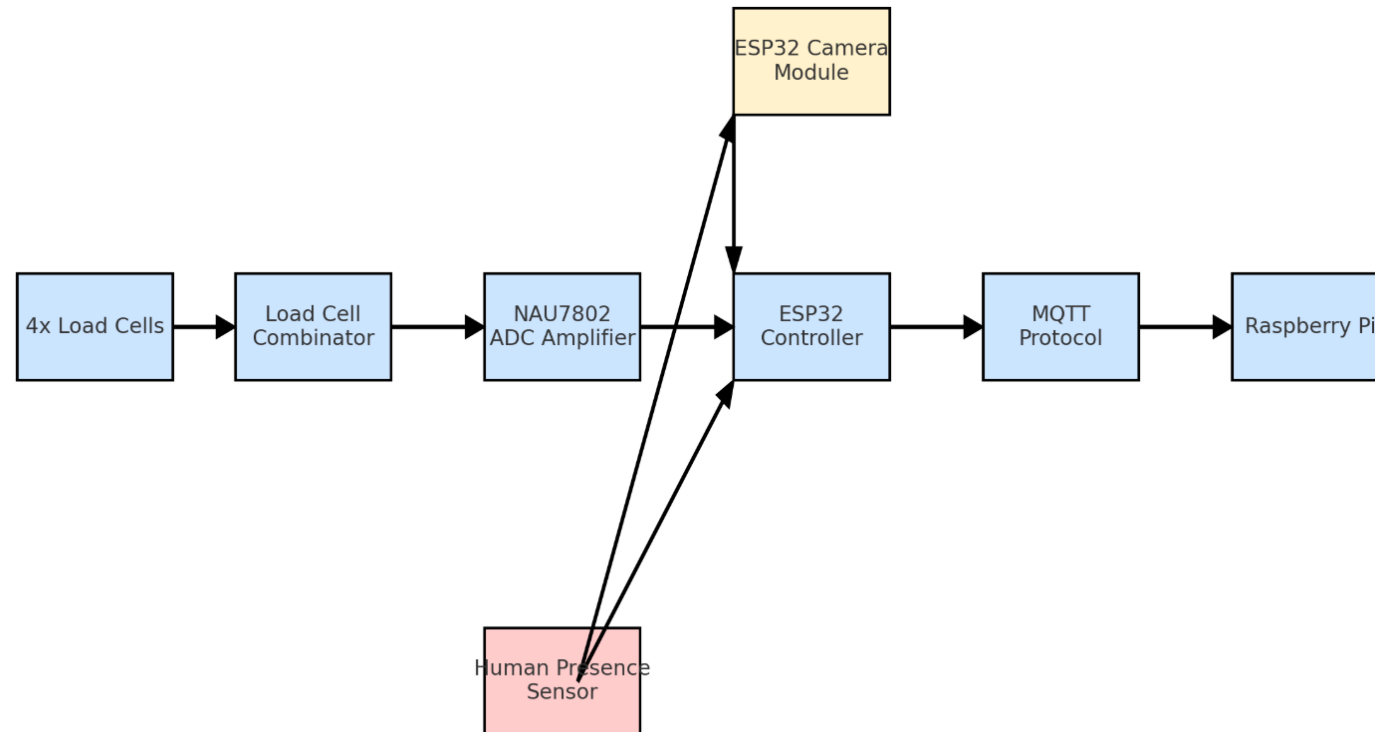


INTRODUCTION

- Goal: build an IoT SmartShelf that:
 - Measures stock in real-time
 - Provides visual verification of shelves
 - Updates inventory automatically in Odoo ERP
 - Operates efficiently (presence-based activation)
- Workflow Diagram:
 - Customer interaction → presence detected → ESP32 wakes sensors
 - Load cell measures weight → ESP32 calculates weight
 - Camera captures shelf image
 - Data sent → Raspberry Pi
 - Raspberry Pi updates Odoo ERP + logs data in InfluxDB/Grafana

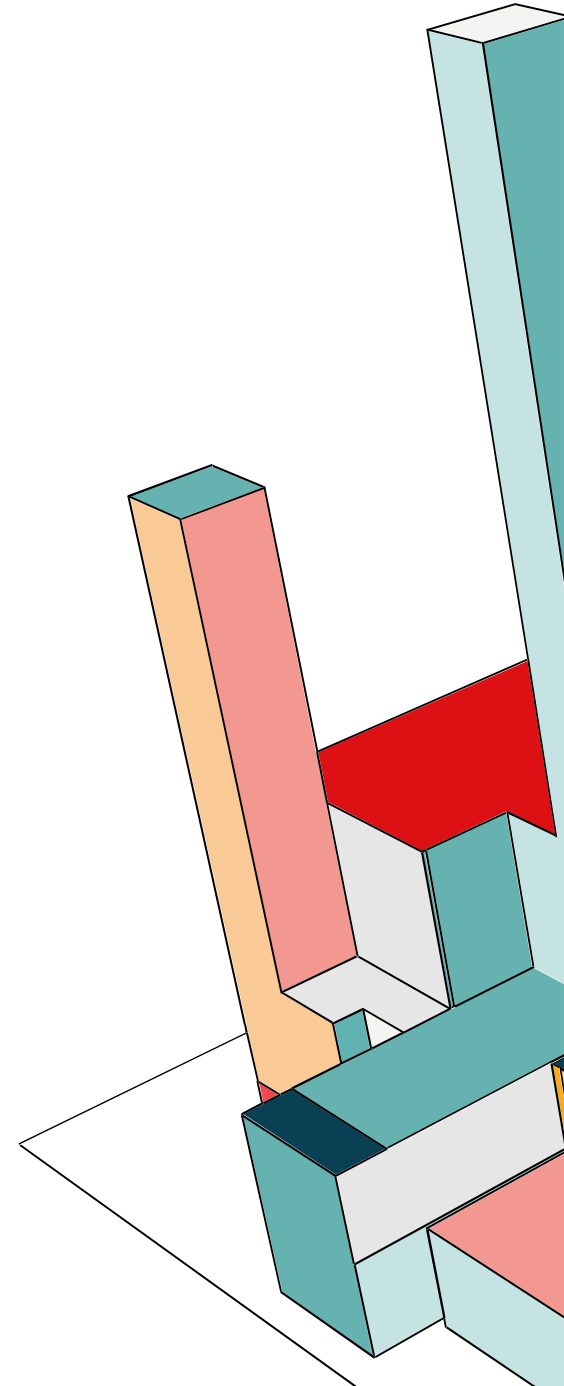


SYSTEM OVERVIEW



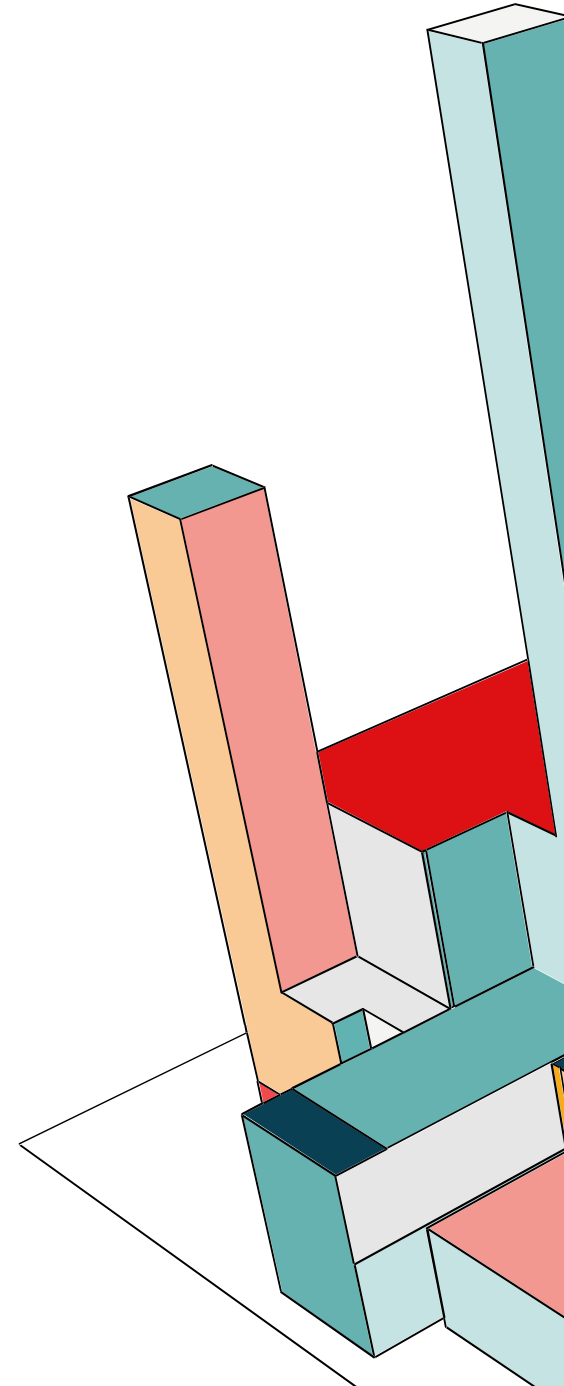
HARDWARE SETUP

- Load Cells (x4) → combined via load cell combinator → NAU7802 ADC Amplifier
- Human Presence Sensor → triggers measurements & camera
- ESP32-S3:
 - Scale + sensor (MQTT)
 - Camera (HTTP/MQTT)
- Raspberry Pi: Wi-Fi, MQTT broker, Flask server



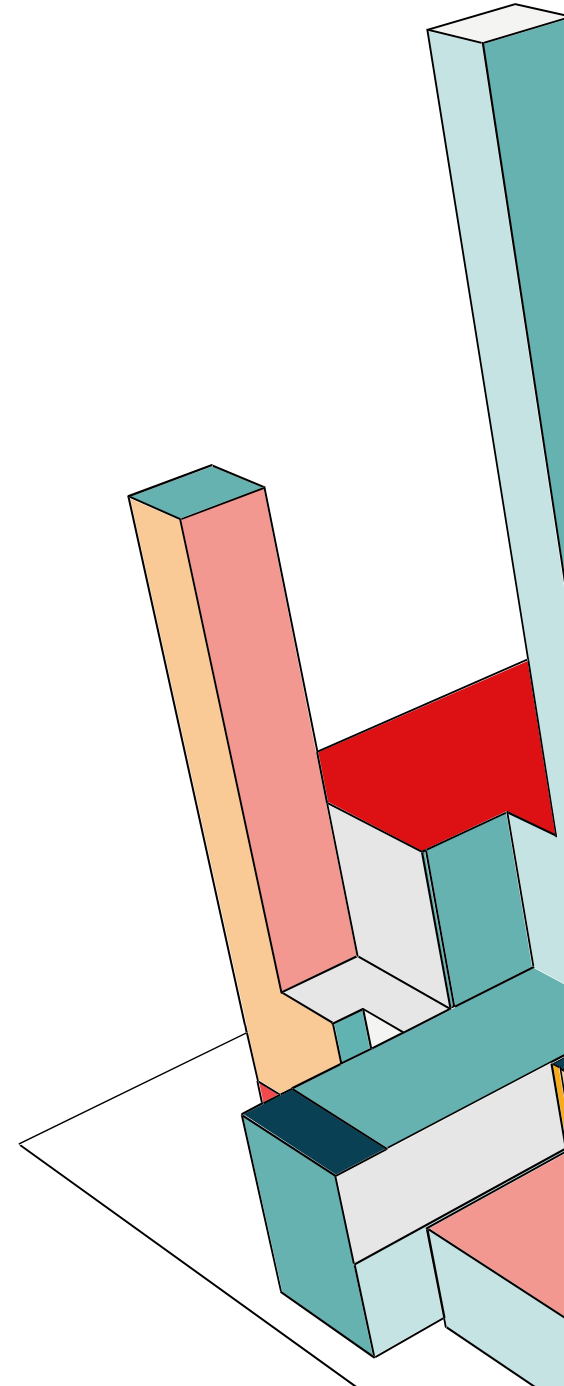
SOFTWARE DEVELOPMENT

- Microcontroller unit (MCU) Firmware:
 - Weight readings → MQTT
 - Camera snapshots → MQTT/HTTP
 - Power management
- Raspberry Pi Software (Python):
 - MQTT client + Flask server
 - Tkinter GUI for real-time display
 - Odoo XML-RPC API for ERP sync
 - Grafana dashboards for visualization



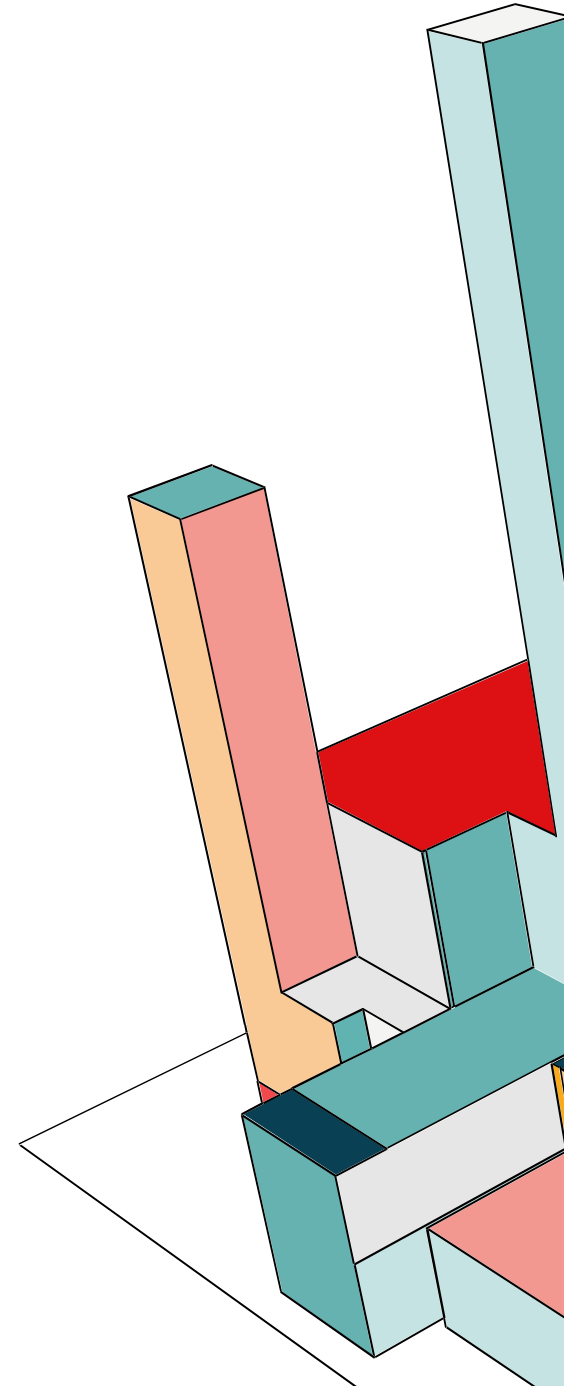
ODOO ERP INTEGRATION

- MQTT data (Scale/Camera) → Python script → Excel → Odoo update
- Features:
 - Fetch product weights dynamically from Odoo
 - Convert weight → item counts
 - Update stock levels in ERP automatically
 - Low-stock alerts & logging



GUI & VISUALIZATION

- Tkinter GUI (Planogram):
 - Shows shelf product, item count, movement detection, latest image



GUI & VISUALIZATION

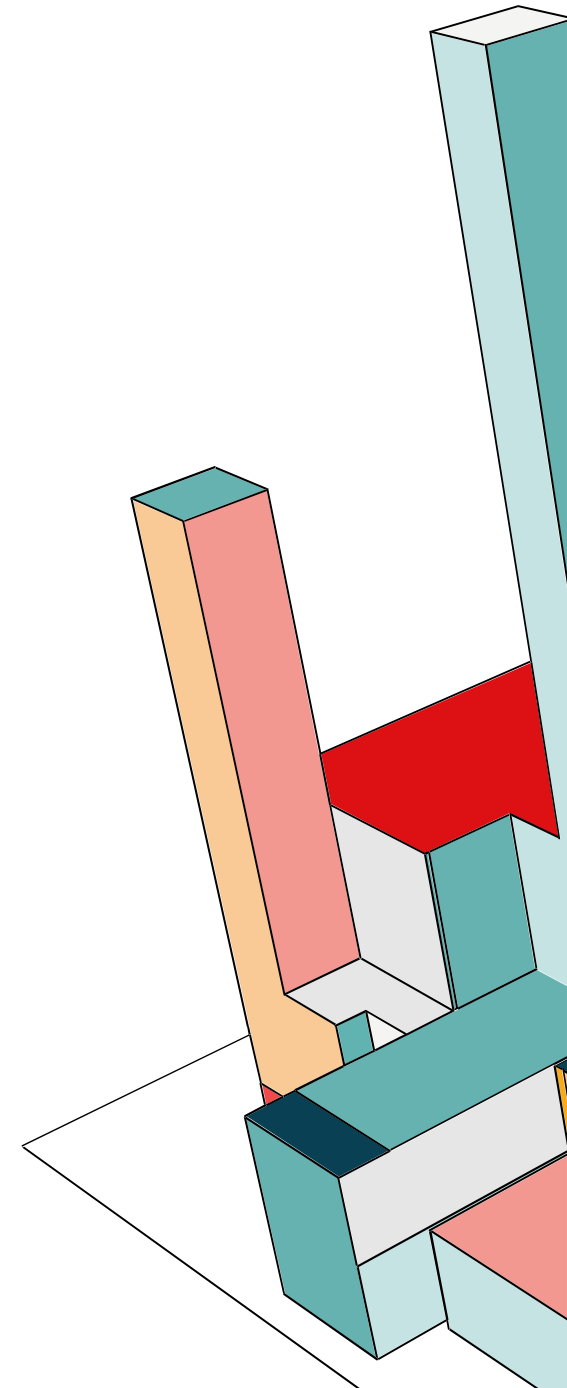
Grafana Dashboard:

- Live weight monitoring
- Time-series trends
- Stock count panels



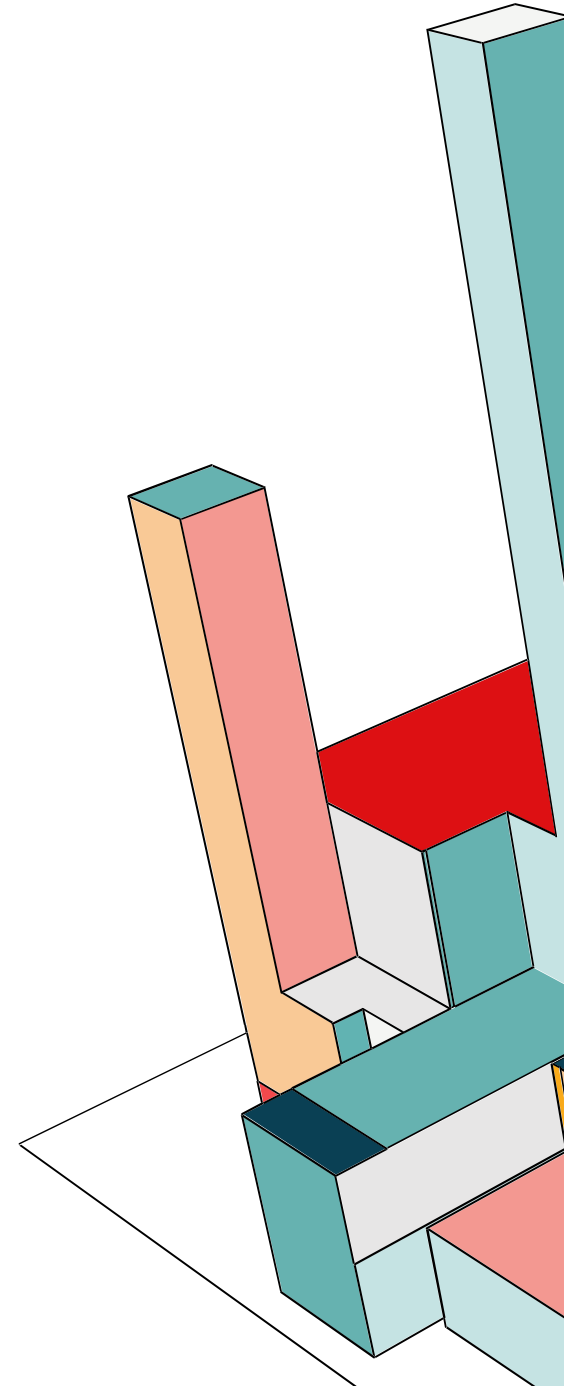
CHALLENGES & SOLUTIONS

- Negative weight values → fixed by correct wiring
- Load-cells noise → Low pass filter
- Images are not displayed quickly/lost → resolved with Flask HTTP server
- Odoo connection errors → fixed with barcode formatting + caching



RESULTS & ACHIEVEMENTS

- ☒ End-to-end pipeline working:
 - ESP32 → MQTT → Raspberry Pi → Odoo → GUI
- ☒ Automatic Odoo updates tested successfully
- ☒ Camera images + sensor events synchronized
- ☒ Real-time GUI and Grafana dashboards are functional
- ☒ Fast updates: full data cycle completes in just a few seconds



CONCLUSION

Scalability

- System can easily expand to multiple shelves and locations.
- Flexible architecture supports both MQTT and HTTP protocols.
- Odoo integration allows seamless adoption in real retail setups.

Innovation

- Unique combination of IoT + ERP + GUI visualization.
- Provides both quantitative data (weights) and qualitative data (images).
- Creates a foundation for future AI-driven shelf monitoring.

CONCLUSION

Operational Impact

- SmartShelf provides automated, real-time inventory tracking.
- Reduces manual stock checks and human errors.
- Ensures data consistency between shelves and Odoo ERP.

System Reliability

- Combined weight + image verification increases accuracy.
- Presence-triggered measurements reduce power usage and noise.
- Modular design ensures each shelf works independently.

THANK YOU

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